

Hyewon Park

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Education

Integrated MS/PhD	Gwangju Institute of Science and Technology, Mechanical and Robotics Engineering	Mar 2023 – Present
BS	Ulsan National Institute of Science and Technology, Mechanical and Aerospace Engineering	Mar 2012 – Feb 2018

Research Interests

- **Goal:** Developing human-joint impedance estimation methods adaptable to diverse rehabilitation robots, enabling personalized therapy
- Estimation of human joint impedance (stiffness, damping, and inertia) in upper-limb application
- Identifying time-varying impedance parameters with deep-learning models (e.g., Bi-LSTM)
- Integrating impedance estimators into rehabilitation-robot control systems

Work Experience

LG Electronics, Air Solution R&D Lab, Research Associate	Changwon, South Korea Dec 2017 – Sep 2020
• Participated in the development of heat pump water heaters for North American and European markets • Collaborated on the development of core electrical components, including electric heaters and powered anode rods. • Contributed to the establishment of reliability test standards and jointly conducted component/system-level testing • Supported UL certification and led CE compliance testing for product safety approval • Authored technical documentation including user manuals, service manuals, BOMs, and exploded views	

Conference

Hyewon Park, Pilwon Hur, *Bi-LSTM based Estimation of Time-Varying Joint Impedance: Comparison with Classical Methods Using Simulation*, ICROS, June 25-27, 2025, Jeonju, South Korea

Hyewon Park, Pilwon Hur, *Human joint stiffness estimation based on machine learning to control mobile CPM(Continuous Passive Motion) machine*, ICROS, July 2-4, 2024, Daejeon, South Korea

Research Projects

Translational Research on Modular Upper Limb Rehabilitation Robot for Improving Upper Limb Function in Patients with Brain Lesions	2025 - Present
• Funding Agent: National Rehabilitation Center, Ministry of Health and Welfare of South Korea • Estimated patient joint impedance (spasticity and stiffness) using deep learning techniques • Led the project and coordinated overall research and development	
Development of a Portable Continuous Passive Motion (CPM) Device	2023 - 2024
• Funding: Sunhan Hospital • Designed and developed a mobile application for user interface control	

- Assisted in hardware development, including strap design and modular motor attachment /detachment

Patent

Yose Bae, Minsu Kim, **Hyewon Park**, *Water heater and controlling method for the same*, KR102382516B1, LG Electronics, granted on April 5, 2022.

Teaching & Assistant Experience

Summer Motor Boot Camp - Planner & lead instructor

2024 Fall

Signals and Systems in Mechanical Eng. - Teaching Assistant, monthly recitations & grading

2023 Spring